

Prof. Dr.-Ing. Jan-Niklas Voigt-Antons

Professor of Computer Science (Immersive Media) · Hamm-Lippstadt University of Applied Sciences
Director, Immersive Reality Lab · Guest Researcher, Technische Universität Berlin

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€4.609 M

third-party funding

234

publications

2,987

citations, h-index 29

ITU-T P.812

co-developer

I study how virtual and augmented reality systems can be built so they actually work for people. My work pairs EEG, eye tracking and physiological sensing with XR deployments in hospitals, public space and safety-critical training — measuring what questionnaires cannot capture.

APPOINTMENTS

since 2024	Co-Head, Computer Visualistics and Design degree programme (with a co-lead), HSHL
since 2024	Founder and Director, HealthTech-HL research group
since 2022	Member of the Commission for Quality Improvement, HSHL
since 2022	Lead, Human-Machine Interaction research area , HSHL
since 2022	Member of the Department Council, HSHL
since 2021	Professor of Computer Science (Immersive Media) , Hamm-Lippstadt University of Applied Sciences · Director, Immersive Reality Lab
since 2021	Guest Researcher, Technische Universität Berlin
2018–2022	Guest Researcher, DFKI Berlin
2015–2021	Senior Research Scientist, Technische Universität Berlin
2012	Guest Researcher, INRS Montreal
2009–2014	Research Scientist, Telekom Innovation Laboratories

EDUCATION

2014	Dr.-Ing. (Doctor of Engineering), Technische Universität Berlin, Faculty IV — Electrical Engineering and Computer Science, <i>magna cum laude</i> . Dissertation: <i>Neural Correlates of Quality Perception for Complex Speech Signals</i> , published as a monograph by Springer.
2008	Diplom-Psychologe (Psychology), Technische Universität Darmstadt, overall grade <i>sehr gut</i> .

AWARDS & RECOGNITION

2025	Best Paper Award , MMVE Workshop @ ACM MMSys
2024	Key Innovator , Innovation Radar of the European Commission — two innovations in XR digital twins
2024	Special Prize, Sustainability Award of Volksbank Beckum-Lippstadt (for a supervised Bachelor's thesis)
2023	Best Journal Paper Award , IEEE Communications Society (MMTC)
2023	Diversity and Societal Impact Award, QoMEX
2021	Conference Paper Honorable Mention, IEEE VR

BIBLIOMETRICS

2,987 citations · h-index 29 · i10-index 83 (Google Scholar, 2 August 2026)

1,662 citations · h-index 22 (Scopus)

234 publications: 2 monographs / edited volumes · 5 book chapters · 45 peer-reviewed journal articles · 163 peer-reviewed conference papers · 17 contributions to international standardization · 2 VDE position papers

STANDARDIZATION

Co-developer of **ITU-T Recommendation P.812**, *Principles of subjective test methods for interactive virtual reality (VR) applications* (2024)

17 contributions in total: ITU-T Study Group 12 (P.812, P.IntVR, P.PHYSIO, G.1035, P.GAME)

ISO/IEC JTC 1/SC 29/WG 2 (MPEG) — Metaverse use cases

DKE at DIN and VDE — digital qualification in healthcare

THIRD-PARTY FUNDING

Approximately **€4.609 million** in continuous third-party funding since 2015, across **22 funded projects** from 10 funding lines, with awards running through 2029

20 of 22 projects (€4.509 million) applied for and led independently; 3 coordinated as consortium lead

Funders: European Union (Horizon Europe, EIT Digital, Joint Programming Initiative), State of North Rhine-Westphalia (EFRE), BMBF, BMW/BMWK, BMDV, Innovation Fund of the Federal Joint Committee (G-BA), GKV-Spitzenverband, DAAD, industry partners

Period	Project	Funder	Role	Volume
2026–2029	ITT	EFRE	Subproject lead	€619k
2026–2029	XRT-HuFuSa	EFRE	Consortium coordinator	€289k
2025–2027	Silent Bed Monitor	ZIM	Subproject lead	€218k
2025–2027	XRwise	Innovationsprogramm für Geschäftsmodelle und Pionierlösungen	Subproject lead	€130k
2024–2026	ARiadne	ZIM	Subproject lead	€208k
2024–2025	Fast Jets	Training Optimization Programme	Sole applicant	€44k
2023–2027	Digitalise SÜDWESTFALEN	Innovative Hochschule	Subproject lead	€290k
2023–2025	DIDYMOS-XR	HORIZON Action Grant	Subproject lead	€303k
2023–2025	DigiOnTrack	mFUND	Consortium coordinator	€296k
2023	Body Interaction Lab	HSHL	Sole applicant	€246k
2022–2025	MIA-PROM	KIAS	Subproject lead	€206k
2020	ART	EIT Digital	Subproject lead	€299k
2020	BGF Pflege	Charité / AOK Nordost	Co-applicant	€39k
2019–2020	Mobil im Havelland	Charité / BMBF	Co-applicant	€61k
2019	Physiological workload measurement	Huawei	Sole applicant	€194k
2019	The health score	e-Therapists GmbH	Sole applicant	€20k
2018–2021	DemTab	Innovationsfonds	Subproject lead	€501k
2018–2021	VoiceAdapt	JPI More Years, Better Lives	Consortium coordinator	€256k
2018	GESOBAU workshops	GESOBAU AG	Sole applicant	€9k
2016	Augletics	Augletics GmbH	Sole applicant	€3k
2016	Post-doc fellowship	TU Berlin	Sole applicant	€74k
2015–2018	PflegeTab	GKV-Spitzenverband	Subproject lead	€304k
22 projects				€4, 609k

SUPERVISION

5 ongoing doctoral projects in the Immersive Reality Lab, spanning: Quality of Experience of medical applications with virtual and physical assistants; psychophysiological measurement of stress and cognitive load in XR; effects of immersive technologies in healthcare and psychotherapy; augmented reality navigation and user experience; affective computing and emotion recognition in XR

4 completed doctoral examinations supervised or reviewed at TU Berlin, including one binational cotutelle with the University of Technology Sydney

35 Master's and 70 Bachelor's theses supervised since 2011

TEACHING

Continuous teaching activity **since 2013**, at Technische Universität Berlin and Hamm-Lippstadt University of Applied Sciences, in German and English

Topics: Augmented Reality · Virtual Reality · Interactive Systems · Graphical User Interfaces · UX Research · Software Ergonomics · Visual Computing · Ubiquitous Computing · Game Development · Introduction to C and C++ · Usability Engineering · Physiological Computing

Module responsibility: *Human Perception* („Mensch und Wahrnehmung“) and *UX Research*

SERVICE & LEADERSHIP

Leadership

Director, Immersive Reality Lab, Hamm-Lippstadt University of Applied Sciences

Mission: developing and evaluating trustworthy XR systems for learning, work and public-space interaction

Team: **11 research staff and 7 student assistants**; annual budget of approximately €1 million

Infrastructure: XR headsets, eye-tracking setups, physiological sensing, behavioural observation and coding

Community & editorial

since 2026 Steering Committee, International Conference on Quality of Multimedia Experience (QoMEX)

2026 Technical Program Chair, MMVE 2026, Hong Kong (co-located with ACM MMSys '26)

2022 General Chair, QoMEX 2022, Lippstadt

since 2025 Editorial Board, *Virtual Worlds* (MDPI)

since 2021 Editorial Board, *Quality and User Experience* (Springer)

2025–2027 Advisory Board, TD-CHAT (Charité Berlin)

2023–2025 Advisory Board, STOP-CSAM (Charité Berlin)

Reviewing & programme committees

Conferences: ACM CHI · IEEE ISMAR · ACM VRST · QoMEX · INTERSPEECH · ICASSP

Journals: IEEE Transactions on Visualization and Computer Graphics · IEEE Transactions on Multimedia · JMIR · Experimental Brain Research · Journal of Building Engineering

Invited talks at industry events, universities and international venues

SELECTED PUBLICATIONS

Chosen by a fixed rule rather than by preference: every publication carrying an award, then the most recent journal articles. The complete list of 234 — with filters by year, type and topic — is at voigt-antons.de/publications/.

[J33] Da Silva, D. A., Da Silva, A. S., Lima, D. D., Da Costa, J. P., De Melo, L. O., Miranda, C., Santos, G. A., Vinel, A., Mendes, P., Verhoeven, S., Voigt-Antons, J.-N. & De Freitas, E. P. (2026). **Spoof detection framework for V2X systems via tensor-based DoA estimation and Yolo-based object detection**. IEEE Access.

[J34] Hinzmann, S., Bestgen, A.-K., Schorlemmer, J., Beierlein, C., Kolbe, J. & Voigt-Antons, J.-N. (2026). **A Modular Questionnaire for Target-Group-Specific Evaluation of Event Formats: Developed in the Context of Virtual Worlds Knowledge Transfer**. *Virtual Worlds*, 5(1), 10.

[J35] Peperkorn, N. L., Ohse, J., Fox, J., Limberg, S., Hadžić, B., Rättsch, M., Voigt-Antons, J.-N., Witthöft, M. & Shibani, Y. (2026). **Effects of Change in Dysfunctional Beliefs and Self-Esteem in Avatar-Based Cognitive Therapy for Symptoms of Social Anxiety Disorder: A Randomized Parallel Trial**. *Scientific Reports*, 16, 6144.

[J36] Ashrafi, N., Graf, P., Marquardt, M., Harnisch, P., Hillmann, S., Ploner, N., Compagna, D., Cirit, E., Papst, L. & Voigt-Antons, J.-N. (2026). **Multimodal Assistance in Rehabilitation: User Experience of Embodied and Non-Embodied Agents for Collecting Patient-Reported Outcome Measures**. *Virtual Worlds*, 5(1), 15.

[J37] Haeger, C., Mümken, S. A., Spang, R. P., Brauer, M., O'Sullivan, J. L., Lech, S., Xue, Q.-L., Stockburger, M., Keller, J., Voigt-Antons, J.-N. & Gellert, P. (2026). **Out-of-home mobility enhancement by a physiotherapist-led motivational**

- counseling intervention among rural community-dwelling older adults 75+: The MOBILE RCT.** BMC Geriatrics, 26, 481.
- [J38] Henning, J., Hinzmann, S., Vona, F., Amer, M. & Voigt-Antons, J.-N. (2026). **Hearing the Way: Influence of Varying Spatialized Audio Cues and Device Type on AR Navigation Experience.** IEEE Transactions on Visualization and Computer Graphics, pp. 1–8.
- [J39] Hallmann, M. C., Peperkorn, N. L., Vona, F., Shibani, Y. & Voigt-Antons, J.-N. (2026). **User Experience and Therapist Effectiveness of Different Virtual Character Types in Virtual Reality Exposure Therapy: A Mixed-Methods Study.** JMIR XR Spatial Comput. — *In press*
- [C90] Kojic, T., Vergari, M., Warsinke, M., Ali, D., Möller, S. & Voigt-Antons, J.-N. (2025). **Multimodal User Experience in Extended Reality: Exploring Hand Tracking, Voice, and Passthrough Interactions.** Paper presented at the international workshop on Immersive Mixed and Virtual Environment Systems (MMVE 2025), Stellenbosch, South Africa. — *Workshop Best Paper Award, MMVE 2025*
- [J32] Antoun, M., Elhajj, I. H., Papadopoulos, P. K., Rizk, A., Islam, T., Huerta, I., Toledo, L., Arvanitis, G., Moustakas, K., Jevtić, A., Agustí, I., Vona, F., Voigt-Antons, J.-N., Warsinke, M., Bailer, W., Thallinger, G., Howkins, B., Poder, I., Ruess, M. & Asmar, D. (2025). **Interactive digital twins enabling responsible extended reality applications.** Scientific Reports, 15, 34539.
- [OJ6] Kuhlencord, M., Ohse, J., Fox, J., Peperkorn, N., Rättsch, M., Voigt-Antons, J.-N. & Shibani, Y. (2025). **Der Einsatz von vielversprechenden Technologien in der klinisch-psychologischen Diagnostik: Fokus auf KI und XR.** Psychologie in Österreich, (4 & 5), 278-285.
- [C71] Ashrafi, N., Schmitt, V., Spang, R. P., Möller, S. & Voigt-Antons, J.-N. (2023). **Protect and Extend - Using GANs for Synthetic Data Generation of Time-Series Medical Records.** Paper presented at the 15th International Conference on Quality of Multimedia Experience (QoMEX 2023), Ghent, Belgium. — *Diversity and Societal Impact Award, QoMEX 2023*
- [C66] Vergari, M., Kojić, T., Vona, F., Garzotto, F., Möller, S. & Voigt-Antons, J.-N. (2021). **Influence of Interactivity and Social Environments on User Experience and Social Acceptability in Virtual Reality.** Paper presented at the IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 2021), Lisbon, Portugal. — *Conference Paper Honorable Mention, IEEE VR 2021*
- [J18] Porcu, S., Floris, A., Voigt-Antons, J.-N., Atzori, L. & Möller, S. (2020). **Estimation of the Quality of Experience During Video Streaming From Facial Expression and Gaze Direction.** IEEE Transactions on Network and Service Management, vol. 17, no. 4, pp. 2702-2716. — *Best Journal Paper Award, IEEE Communications Society (MMTC) 2023*
- [J6] Engelke, U., Darcy, D. P., Mulliken, G. H., Bosse, S., Martini, M.G. Arndt, S., Antons, J.-N., Chan, K. Y., Ramzan, N. & Brunnström, K. (2017). **Psychophysiology-based QoE Assessment: A Survey.** Journal of Selected Topics in Signal Processing, 11, 6-21. — *Most-cited work*

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